Neuroimaging Correlates of H3K27M-altered Diffuse Midline Glioma
A Single Entity with Many Faces: Systematic Review and Meta-analysis

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INTRODUCTION

- In 2016, the World Health Organization introduced for the first time a separate entity of "H3K27M-mutant" diffuse infiltrative midline glioma in the classification of CNS tumors. These mutant lesions exhibit aggressive clinical behavior with significantly worse prognosis and survival rates than wild-type counterparts. They are designated as grade-IV independent of their histologic features. Imaging features of these tumors are overlapping (figure 1), with little known regarding the neuroimaging correlates of their mutational status.
- Histone (H3) has three main variants; H3.1 and H3.2, which are deposited in chromatin during DNA replication, and H3.3 (not related to DNA replication and found in heterochromatin). Although the diagnosis has been relied widely on imaging and clinical findings for years due to location-challenging tissue biopsy, improved stereotactic techniques have made the procedure generally safe and more accurate.
- The aim of this paper is to review the current literature for potential neuroimaging correlates of H3K27M mutational status in diffuse midline gliomas. To the best of our knowledge, the literature has no published systematic reviews/meta-analyses on neuroimaging in H3K27M-altered diffuse midline glioma.

METHODS

A systematic search of PubMed, Scopus, and Embase databases, was conducted to identify relevant studies between 01/01/2017 and 15/12/2021 (PRISMA flow chart illustrated in figure 1). Two independent raters used the Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2) for evaluation of the quality of studies under review (figure 2). Beside descriptive results, random-effects models for odds ratio and standardized mean difference meta-analysis were employed to assess the difference between H3K27M-mutant versus wild-type groups in terms of tumor location, contrast enhancement and diffusion characteristics.

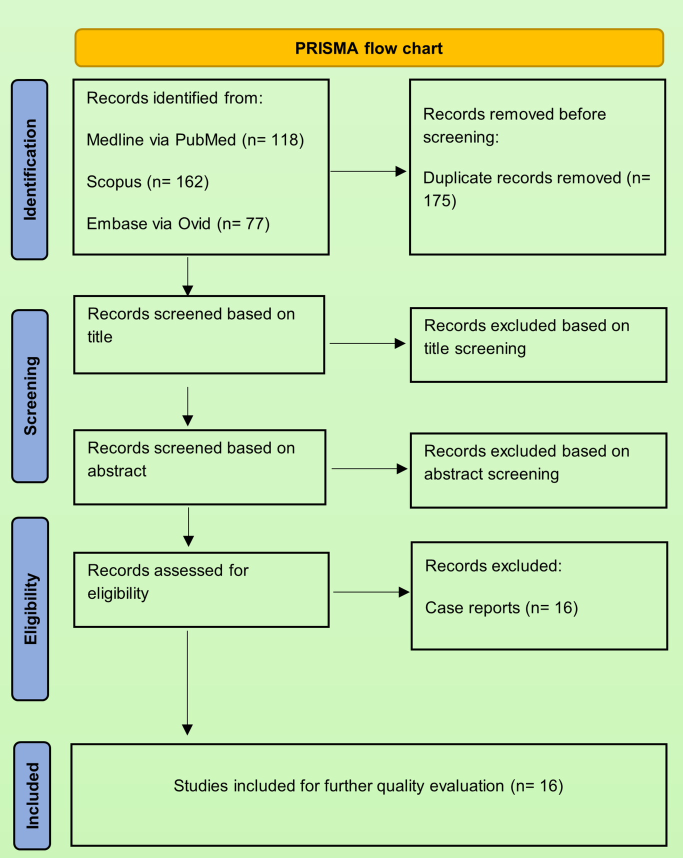

Figure 1: PRISMA flow chart of the steps followed during article selection

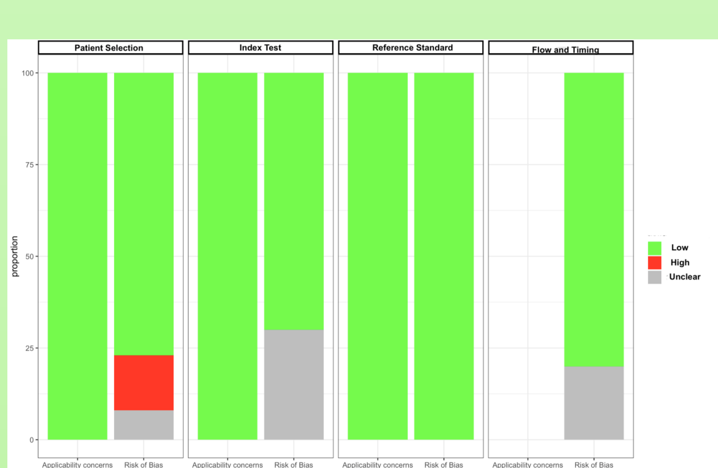

Figure 2: QUADAS-2 graphic showing quality assessment of the 16 papers, 3 of them were excluded due to risk of bias in both patient selection and index testing domains. Applicability concerns were at low risk level across patient selection, index testing, and reference standard domains.

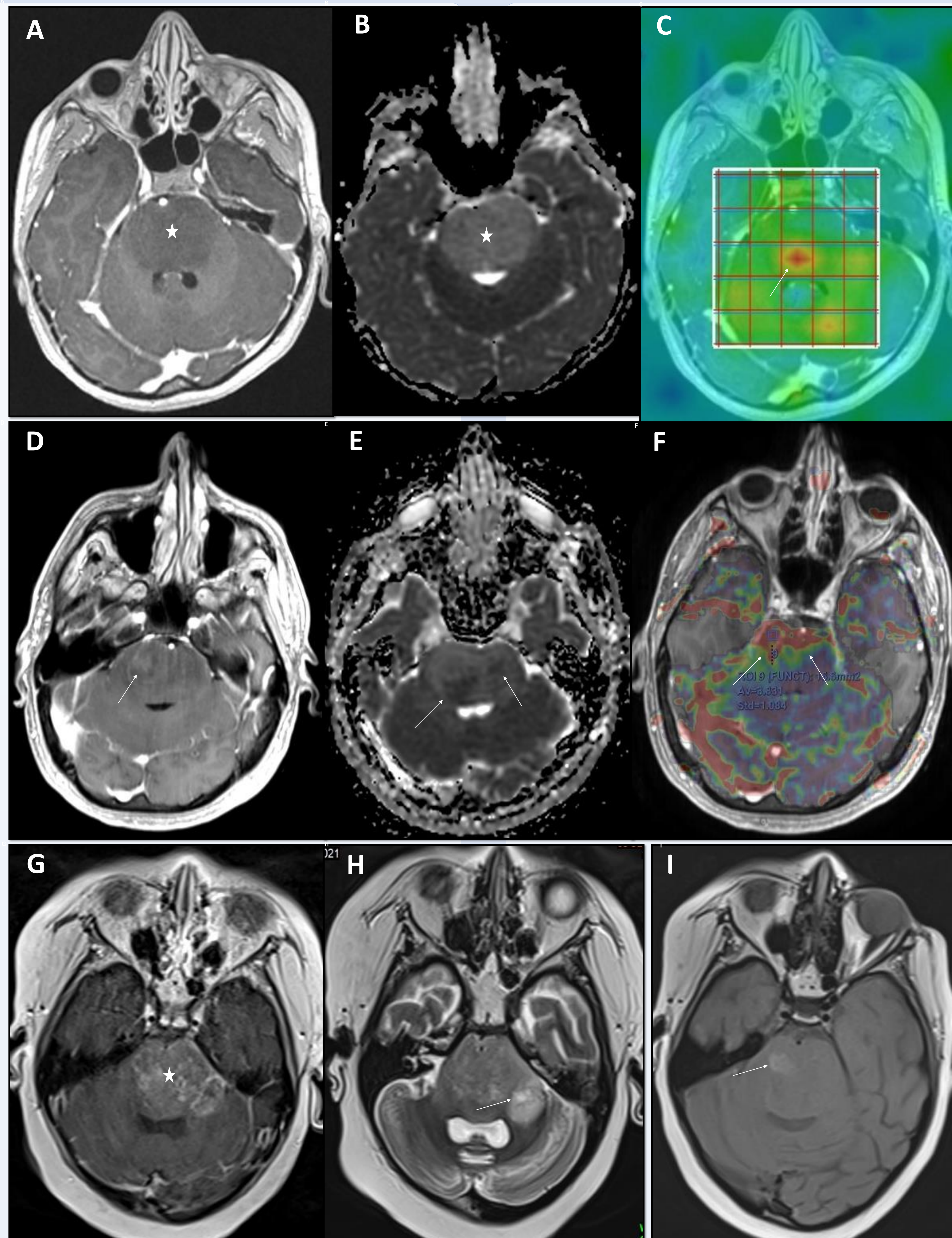

Figure 3: MR studies in three patients with diffuse pontine gliomas from the archives of our institution. Patient 1 (top [A-C]): Non-enhancing diffuse pontine mass lesion (star in A), without diffusion restriction on ADC map (star in B), but a central core of elevated Choline/NAA ratio on multi-voxel spectroscopy (arrow in C). Patient 2 (middle [D-F]): Faint focal enhancement (arrow in D) in diffuse pontine mass, with lower diffusivity/dark ADC areas (arrows in E), and elevated CBV along its right anterolateral aspect on DSC perfusion (arrows in F). Patient 3 (bottom [G-I]): Heterogeneous diffuse enhancement of pontine mass (star in G), with tiny necrotic spot on T2 (arrow in H) towards left cerebellar extension, and hemorrhagic high signal on unenhanced T1 (arrow in I). H3K27M status was mutant in patients 1 & 3, but wild-type in patient 2.

RESULTS

The total number of cases across the combined 13 studies was 621 midline gliomas. H3K27M mutation was present in 312 (50.2%) of them, while the remaining 309 (49.8%) had wild-type tumors. Eight studies had mixed samples of both pediatric and adult cases. Supratentorial tumors were included in 9/13 studies while 12 studies included infratentorial brain lesions, two out of the 12 studies recruited pontine tumors only. Spinal cord tumors were included in 6/13 studies.

By meta-analysis, infratentorial tumors were more likely to harbor H3K27M mutations (figure 4), with a statistically significant odds ratio of 1.616, 95%-CI [1.016, 2.572] p=0.043 and considerable heterogeneity (I^2=44.15%, p=0.111). Enhancing tumors were more likely to harbor H3K27M mutations (figure 5) with a statistically significant odds ratio of 2.806, 95%-CI [1.363, 5.778] p=0.005 but no heterogeneity (I^2=0.0%, p=0.826). The risk of publication is low for both infratentorial location or contrast enhancement odds ratios.

There was no significant difference between H3K27M-mutant and wild-type gliomas neither when using ADCmin (figure 6) SMD -0.72, 95%-CI [-1.60, 0.15] p=0.104 nor ADCmean (figure not shown) SMD 0.20, 95%-CI [-0.28, 0.69] p=0.401. There was considerable heterogeneity for the ADCmin meta-analysis (I^2=70.09%, p=0.035), but no heterogeneity for the ADCmean meta-analysis (I^2= 0.0%, p=0.937). The risk of publication is low regardless of when looking at the contour-enhanced funnel plots or Egger's test.

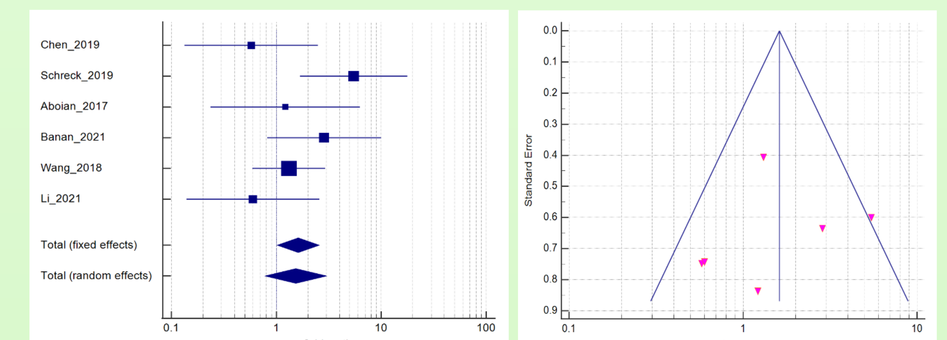

Figure 4: Forest plot (left) and funnel plot (right) for the random-effects model and odds ratio of the infratentorial location in predicting H3K27M-mutant vs H3K27M wild-type midline gliomas.

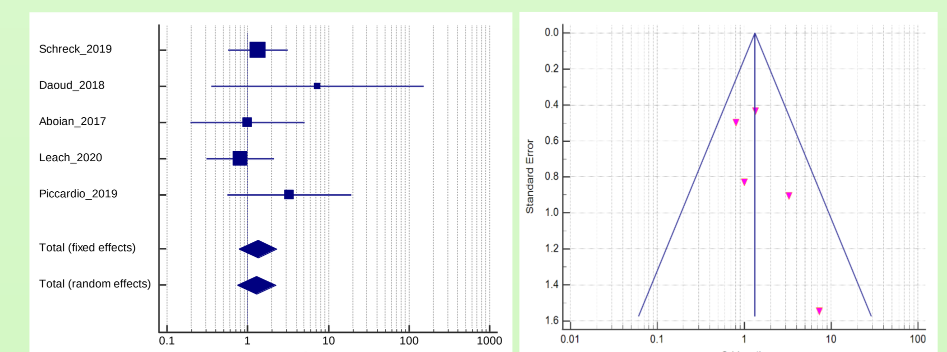

Figure 5: Forest plot (left) and funnel plot (right) for the random-effects model and odds ratio of the presence of contrast enhancement in predicting H3K27M-mutant vs H3K27M wild-type midline gliomas.

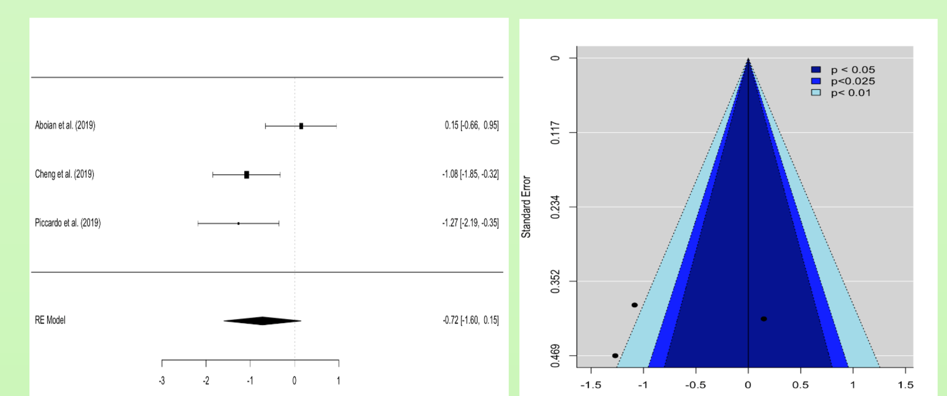

Figure 6: Standardized mean difference ADCmin meta-analysis. Forest plot (left) and contour-enhanced funnel plot (right) for the random-effects model and standardized mean difference ADCmin meta-analysis of H3K27M-mutant gliomas vs H3K27M wild type gliomas.

CONCLUSION

While infratentorial location, contrast enhancement, and other imaging features showed an interesting pattern in the prediction of H3K27M mutational status, the imaging appearance seems to result from a complex interplay of factors including specific location within the midline, histologic grade, and mutational status. Molecular subtypes could also contribute to the variable imaging faces of this tumor entity. Further studies incorporating location-specific differences and secondary molecular profiles are required for accurate reporting of findings.